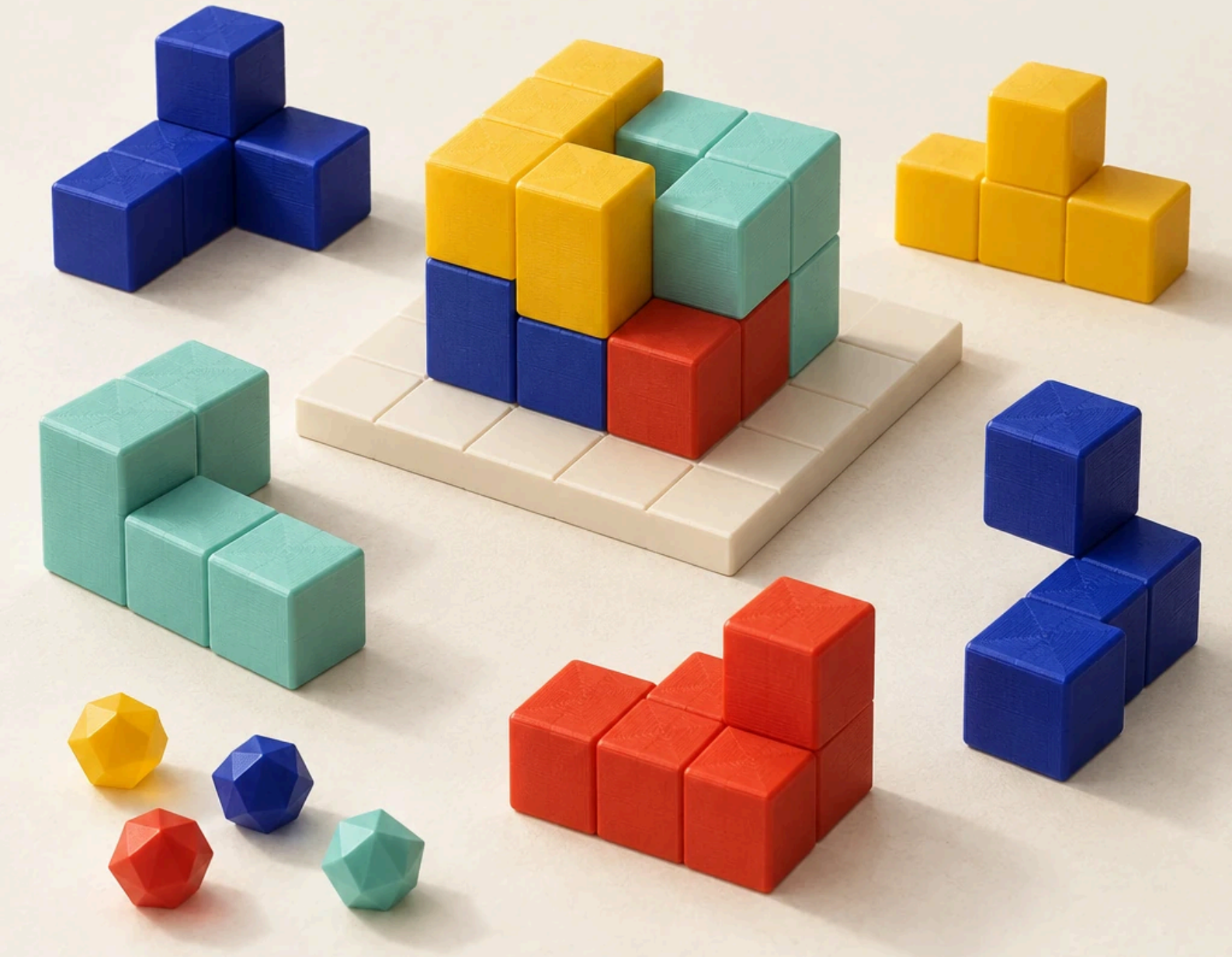
A FAMILY 3D PUZZLE GAME

WONGO

CLASSIC EDITION

Play online
wongo.duras.me



Good things fit together.

Family rulebook · English edition

2–8 players / 9 rounds / 2 layers

WONGO 3D

FAMILY RULEBOOK · ENGLISH · VERSION 1.0

2–8 players · 9 rounds · Ages 10+ suggested · About 25–40 minutes

Build a little solid world before time runs out. Everyone solves a different puzzle at the same time, earns gems, and tries to finish with the most points.

Wongo is an original family adaptation inspired by Ubongo 3D Master. It is not an official Ubongo product. These are the rules for our custom two-to-eight-player game.

What you need

- Copies of the 16 numbered shapes: **40 pieces for 2–4 players, or 80 for 5–8 players**, using the print-manifest quantities. The 80-piece pool also works for 2–4. This is a shared pool, not 16 pieces per player.
- One assigned puzzle per player per round, displayed in the app or printed at the matching piece scale.
- A timer, an opaque gem bag, and paper for recording results.
- **90 gems:** 18 red, 27 blue, 18 green and 27 amber. This one set works for two through eight players.

Gem	Points
Red	4
Blue	3
Green	2
Amber	1

Set up

1. Choose the number of players, from two to eight. Use the round generator so that all assigned puzzles together fit the available piece inventory. Do not independently pick arbitrary puzzles from the shared pool.
2. Agree on difficulty before starting. **Easy uses three pieces in three different colours; hard uses four pieces, one of each colour.** No puzzle uses the same colour twice. These labels describe piece count, not a guarantee of how difficult a person will find a particular puzzle. The initial round planner uses one common level. As a family variant, players may use different levels only if the combined piece allocation has been checked against the shared inventory.
3. Set the timer to **two minutes per round**. You may agree on a longer time for the whole game. Give beginners a practice round first.
4. Put **nine blue and nine amber gems** in two visible rows. Each pair marks one round. Put all remaining gems in the bag: initially **18 of each colour**.
5. Keep solutions hidden. Each player receives a different assigned puzzle and takes exactly the pieces named on it. Take time to identify the pieces before the timer starts.

The building challenge

A puzzle shows a footprint on a square grid and the IDs of the pieces to use. Fill **every footprint square exactly two cubes high**.

- Use every assigned piece exactly once. Use no extra pieces.
- Rotate or turn pieces over in three dimensions. Do not alter or mirror their geometry.
- Leave no gaps in either layer, overlap no cubes, and put nothing outside the footprint or above layer two.
- Build an arrangement that can stand on the table; a hovering or physically impossible placement does not count. You may rearrange or lift pieces while working.

A printed card is the base layer's footprint. The same footprint must be filled again in the second layer. Piece colours and numbers identify shapes; the board's squares are not colour-matching instructions.

Play a round

1. When everybody has the correct pieces, start the timer together.
2. Work on your own puzzle. When finished, say **“Wongo!”** and leave your arrangement in place. Record the finish order; everyone else keeps solving until time runs out.
3. Check completed builds before awarding gems. An incorrect build earns nothing unless the player fixes it and finishes before the timer ends. A displayed solution is not proof of a player's own build.
4. Every player who finished correctly in time draws **one random gem** from the bag. In addition, the fastest receives that round's **blue** gem and the second-fastest receives that round's **amber** gem.
5. Return any unawarded bonus gem from that round to the bag. Return all pieces to the shared pool, then deal the next inventory-checked round.

If two finish announcements cannot be distinguished, draw lots between those players for their order before giving bonus gems. Everyone who solved still gets their random gem.

If **nobody** solves a round, the group may use **one second chance**: restart the same timer once and continue the same puzzles. Agree before the game whether to use this option. If nobody solves after that, award no gems and return both bonus gems to the bag. The round still counts. Never add extra rounds because a round was unsuccessful.

Win

After round nine, add the points on your collected gems. The highest total wins. **A tied highest score is a shared victory.** Gem count alone does not decide the winner.

For mixed abilities, keep each person's chosen level consistent throughout a game. After a game, adjust levels or the common time limit together. The random gem draw deliberately gives slower solvers a chance to win too.

Use the virtual puzzle library

Every puzzle has a permanent **problem code**. Enter or search that code to return to the same footprint, piece list and saved solution. Codes identify Wongo-generated puzzles, not the publisher's original cards.

Choose **Try this puzzle** to build your own solution. Select a piece, rotate it about X, Y or Z, and choose its anchor on the lower or upper grid. The anchor is the lower corner of the piece’s bounding box, which may itself be empty; **+X points right and +Y up the grid**. Place the piece, then repeat. Overlaps and placements outside the two-layer board are rejected. You can undo, remove a placed piece, or reset the board.

During practice, a partly built arrangement may temporarily have unsupported cubes. Completion is checked independently: all assigned pieces must form their correct shapes and exactly fill both layers without gaps or overlaps. Any valid arrangement counts, even if it differs from the saved solution.

To check the reference solution, reveal it, rotate the 3D view, step through the pieces, or inspect either layer. If WebGL is unavailable, use the numbered 2D layer views. Revealing a solution during a competitive round makes that player's puzzle ineligible for rewards that round.

Each published puzzle passes an exact solver check and a separate saved-solution check. The round planner also checks total piece demand. These establish geometric solvability; printed fit still needs a physical sample test.

Practice runs locally in your browser. There is no synchronized online multiplayer: for family games, start a shared timer, record finish order and count gem scores yourselves.

Print later

Print puzzle cards at **100% / actual size**, with page fitting disabled, after choosing your piece size (the initial models use 20 mm cubes). Measure the card's calibration guide before using it with printed pieces. Print solutions separately or on a separate page so they are not visible during play. Start with a small fit test before printing the whole collection.

Solo practice

Choose any puzzle, hide the solution, and time yourself. Try to beat your previous time or solve as many puzzles as possible in ten minutes. Gems and multiplayer inventory checks are unnecessary when solving one puzzle at a time.

Keep small pieces and gems away from children who might put them in their mouths.

Reference: Albi's Ubongo 3D Master rulebook. Wongo's player count, generated puzzles, inventory scheduling and 90-gem setup are custom adaptations.